

Wingspan

A dynamic mid/side width processor that opens the stereo image (or pulls it in) when the music gets loud.



Overview

Wingspan splits the stereo signal into Mid (what is common to both sides - the centre) and Side (the difference - the width). It then turns the Side up or down as the level rises above a threshold, so the image can open out when a chorus lifts, or pull in towards the centre when things get loud and busy.

A detector listens to the Mid, the Side or a blend of the two. When it passes THRESHOLD, the Side moves by AMOUNT: positive amounts widen, negative amounts narrow. The movement is smoothed in decibels and rounded off, so the image glides rather than pumps or clicks. A static BASE WIDTH, a LOW MONO filter that keeps the bass in the centre, and a parallel MIX complete the chain.

The Mid is never touched, so the mono sum stays exactly as it was: Wingspan is mono-safe by design. It has no drift - like the other mastering tools it is deliberately steady - and no latency unless the output stage is set to LIMIT.

Reach for it for

- Choruses that open out wider than the verses
- Reining in spiky, over-wide moments on a master or bus

- A little extra width on a master with the bass kept mono
- Rooms and ambience that bloom with each drum hit
- Keeping a loud lead vocal focused by narrowing the sides around it
- Locking the low end to mono for vinyl, clubs and mono playback

Controls

The header

PRESETS	Opens the preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. A new instance opens in L.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

DYNAMICS

The sidechain that decides how much the Side moves. Each knob shows its value under its name while you hover over it or turn it.

THRESHOLD	The level the detector must pass before the image starts to move, -60 to 0 dB (-20 dB by default). The change starts gently over a 6 dB soft knee.
AMOUNT	How hard the Side is moved above the threshold, -100 to +100% (+30% by default), with a centre detent at OFF. Towards WIDEN the image opens; towards NARROW it closes. At 100% the Side moves by 0.5 dB for every dB over the threshold, up to +12 dB wider or -18 dB narrower.
ATTACK	How quickly the image moves once over the threshold, 0.5 to 100 ms (15 ms by default).

RELEASE	How quickly it settles back afterwards, 20 ms to 2 s (300 ms by default).
DETECT	What the detector listens to, from MID (the centre - vocals, kick, bass) through a blend (M 70 / S 30 and so on) to SIDE (the width itself). MID by default.
PEAK / RMS (DETECTOR)	PEAK reacts to every instant peak; RMS (the default) averages the level over 8 ms, for smoother, more musical movement.
LEVEL vs THRESHOLD	A 24-segment ladder showing the detector level from -60 to 0 dB, with a line at the THRESHOLD. Segments above the threshold light amber.
MOVING	A lamp that lights while the Side is being moved by more than a quarter of a dB.

STEREO FIELD

Displays only, all showing the output.

VECTORSCOPE	The output plotted with Mid upwards and Side across, auto-scaled to the level. A mono signal is a vertical line; the wider the image, the more it spreads sideways. The amber "wings" show the current width angle: straight up is mono.
SIDE GAIN 4 s	The last four seconds of side gain, +12 to -18 dB, newest on the right. Amber above the 0 line is wider, cream below it narrower.
CORRELATION	-1 to +1. Near +1 is close to mono, around 0 is wide; it turns red below 0, where the sides are working against each other and the mono sum loses level.
SIDE GAIN readout	The side gain right now, in dB.
IMAGE WIDTH readout	The resulting width: BASE WIDTH with the current side gain added, as a percentage (100% = as it came in).
Gain-change bar	The same side gain as a bar from -18 (NARROWER) to +12

(WIDER), with 0 in the middle.

WIDTH & OUTPUT

BASE WIDTH	A static width, 0 to 200% (100% by default, with a detent there). 0% is mono, 100% leaves the image as it is, 200% doubles the Side.
PRE / POST (WIDTH STAGE)	Where BASE WIDTH sits. PRE: before the detector, so a detector listening to the Side hears the widened image. POST (the default): after the dynamic stage, so the detector hears the original image. With DETECT on MID it makes no difference.
LOW MONO OFF / 80 / 120 / 200	Keeps the bass mono below 80, 120 or 200 Hz by filtering it out of the Side with a steep 24 dB per octave filter (OFF by default). Switching it fades in and out without a click.
MIX	Parallel blend with the untouched input, 0 to 100% (100% by default).
OUTPUT	Output level, -12 to +12 dB (0 dB by default).
OFF / CLIP / LIMIT (OUT STAGE)	The final stage: nothing, a soft clipper that holds the peaks at -0.3 dBFS (no latency), or a lookahead limiter that holds them at -0.3 dBFS (3 ms latency, compensated by REAPER). OFF by default; the CLIP key flashes red while it is clipping.
IN / OUT meters	Left and right peak meters for the input and the output, -48 to +3 dB; segments above 0 dB light red.

Knobs: drag up or down (Shift for fine moves) or use the mouse wheel; double-click or Ctrl-click to reset a knob to its default.

Presets

Presets set everything in the DYNAMICS and WIDTH sections and MIX. Init also resets OUTPUT and the output stage; the others leave them as they are.

Init	The defaults: -20 dB threshold, +30% widening, listening to the Mid in RMS, no low mono.
Wide Choruses	+60% from the Mid with a slow 600 ms release, slightly wider base, low mono at 120 Hz: the image opens as the song lifts.
Tame Wide Peaks	-50% listening to the Side in PEAK with a fast attack: spiky bursts of width are reined in.
Gentle Master Width	+25%, a 30% Side blend in the detector, slow and smooth, base 110% with low mono at 80 Hz.
Mono-Safe Lows	A light +15% with base 115%, and the bass locked mono below 120 Hz.
Drum Room Bloom	+70% in PEAK with a 2 ms attack and MIX at 70%: each hit blooms the room, in parallel.
Pad Spread	Low threshold, slow attack and 1.5 s release, base 130% in PRE, low mono at 200 Hz: the swell spreads further.
Narrow Loud Bits	-60% above -14 dB: loud passages pull towards the centre.
Vocal Centre Lock	-35%, fast: a loud centre (usually the vocal) narrows the sides around it.
Wide When Quiet	Base 125% with -40% above -24 dB: wide verses that steady up when the track gets big.

Tips

- Start with Init and watch LEVEL vs THRESHOLD: set THRESHOLD so only the loud sections light amber.
- Use DETECT on MID to follow the song's energy (vocals, kick, bass); use SIDE to react to the width itself, for example to tame it.
- Keep an eye on CORRELATION: if it heads below 0, back off AMOUNT or BASE WIDTH.

- On a master, turn LOW MONO to 80 or 120 Hz whenever you widen, so the bass stays solid in mono and on vinyl.
- Because the Mid is untouched, a mono fold-down sounds the same whatever Wingspan is doing - only the width changes.

File: gfx-wingspan.jsfx